

Chiral P,N-ligands for the highly enantioselective addition of diethylzinc to aromatic aldehydes

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A new sterically hindered chiral P,N-ligand was synthesized and successfully applied to copper catalyzed asymmetric addition of diethylzinc to aromatic aldehydes. Various aromatic aldehydes can react smoothly to give the corresponding addition products with good to excellent enantioselectivities, which provides a readily accessible method for the preparation of chiral secondary alcohols.

KEYWORDS

addition, asymmetric catalysis, chiral P,N-ligands, copper, zinc

1 | INTRODUCTION

Optically active secondary alcohols are not only versatile building blocks in organic synthesis but also important structural motifs in a number of natural products as well as in pharmaceuticals.^[1] Due to their importance, considerable effort has been made to develop efficient method for the synthesis of chiral secondary alcohols.

The asymmetric addition of organozinc to aldehydes in the presence of chiral ligands is an important method for the preparation of chiral secondary alcohols, and thus has received considerable attention over the past decades.^[2] Furthermore, this reaction is now also regarded as a standard reaction for benchmarking the efficiency of

new chiral ligand. The enantioselectivity in this reaction is mainly depend of the chiral ligand, and therefore, the search of new ligand for this important asymmetric transformation is a field of continuous interest. In 1984, Oguni and Omi^[3] reported the first enantioselective addition of diethylzinc to benzaldehyde in the presence of (*S*)-leucinol, affording the addition product with moderate enantioselectivity (49% ee). Subsequently, Noyori^[4] and coworkers used (–)-3-*exo*-(dimethylamino) isoborneol (DIAB) as the first highly enantioselective catalyst for the dialkylzinc to aldehydes, producing the product in 98% ee. Following these preliminary examples, many types of chiral ligands have been developed and evaluated in the asymmetric addition of organozinc to

aldehydes, giving products with high enantiomeric excesses. These include amino alcohols,^[5] amino thiols,^[6] imino alcohols,^[7] aminol diols,^[8] diols,^[9] diamines,^[10] bissulfonamides^[11] and hydroxyl acids^[12] (Figure 1). Despite these achievements, the ligands types are usually N,O-, N,O,O-, N,S-, N,N-, O,O-donors, the design and development of efficient chiral ligands with novel backbone still remains an enduring interest in asymmetric catalysis.

In the past few decades, the design, synthesis and application of chiral P,N-ligands has emerged as a high active field of research. And the chiral P,N-ligands are now among the most important and widely used ligands for asymmetric catalysis. Because each donor atom of P, N-ligands can be individually modified both sterically and electronically, these ligands have a feature of high fine-tuning. In fact, the chiral P,N-ligands have been found to display wide utilities in various catalytic asymmetric transformations, giving excellent yields and enantioselectivities in hydrogenations, hydroborations, allylic alkylations, conjugated additions and cycloadditions.^[13] However, to the best of our knowledge, there are no reports on the highly enantioselective addition of organozinc to aldehydes with chiral P,N-ligand, although the resulted chiral secondary alcohols are useful intermediates of nature products and pharmaceutical products. Herein, we would like to report the synthesis and application of chiral P,N-ligands (Figure 1) in the copper-catalyzed asymmetric addition of diethylzinc to aldehydes with excellent enantioselectivities.

2 | RESULTS AND DISCUSSION

In 1999, Zhang^[14] reported the synthesis of chiral P,N-ligands **L1** and **L2** and successfully demonstrated their applications in the copper-catalyzed asymmetric conjugate addition of diethylzinc to enones. We envisioned that this catalytic system should also be suitable for the addition of diethylzinc to aldehydes. We then started our investigation

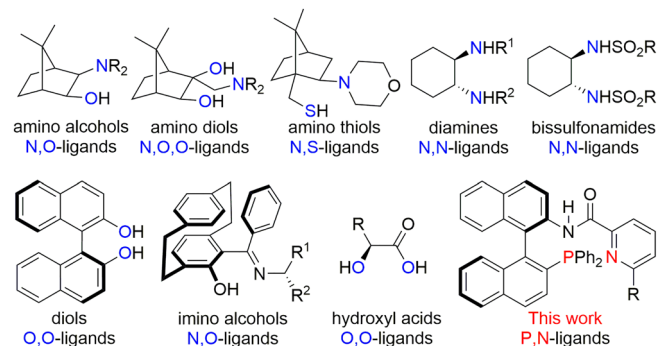
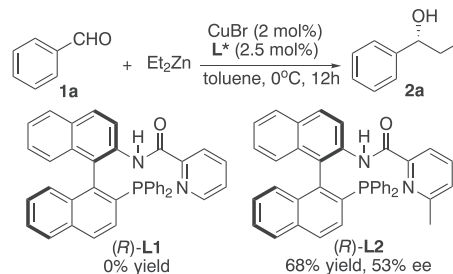


FIGURE 1 Representative ligands used in the asymmetric addition of organozinc to aldehydes

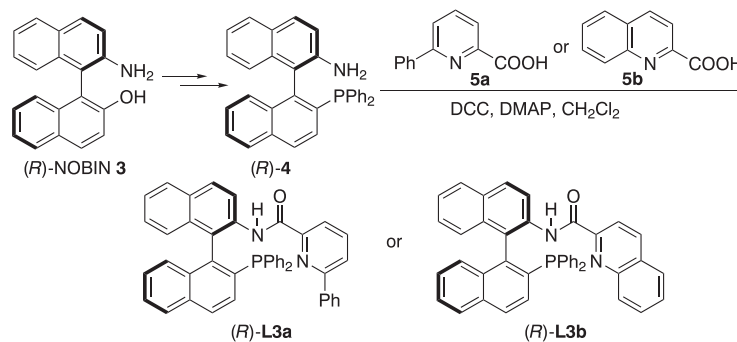
by using benzaldehyde **1a** as the model substrate and the addition reaction was performed at 0 °C in toluene for 12 hours in the presence of 2 mol% catalyst generated *in situ* with CuBr and chiral P,N-ligand (*R*)-**L1**. However, the reaction did not occur at all. Fortunately, the desired addition product (*R*)-1-phenylpropan-1-ol (**2a**) was obtained in 68% yield with 53% ee when 6-methyl pyridinyl substituted ligand (*R*)-**L2** was used (Scheme 1). It's showed that the steric property of ligand has significant effect on the reactivity and enantioselectivity of this transformation. This promising results encouraged us to synthesize the sterically more hindered ligand (*R*)-**L3a** and **L3b** which might be benefit for both the reactivity and enantioselectivity.

The new P,N-ligand (*R*)-**L3a** and **L3b** was prepared from chiral 2-amino-2'-hydroxy-1,1'-binaphthyl (NOBIN **3**) via the procedure as outline in Scheme 2. Thus the (*R*)-NOBIN **3** was converted into (*R*)-2'-(diphenylphosphaneyl)-[1,1'-binaphthalen]-2-amine **4**, which reacted smoothly with 6-phenylpicolinic acid **5a** and quinoline-2-carboxylic acid **5b** in the presence of dicyclohexylcarbodiimide (DCC) and dimethylaminopyridine (DMAP), resulting the targeted P,N-ligand (*R*)-**L3a** and **L3b** in good yield.

With the newly developed chiral P,N-ligand (*R*)-**L3a** and **L3b** in hand, we then investigate its efficiency in the copper catalyzed asymmetric addition of diethylzinc to benzaldehyde **1a** and the results are summarized in Table 1. As expected, excellent result of 90% yield with 96% ee was obtained with 6-methyl pyridinyl substituted ligand (*R*)-**L3a** (entry 1). However, only moderate yield and enantioselectivity was obtained with quinoline-2-carboxylic acid derived ligand (*R*)-**L3b** (entry 2). Encouraged by this result, we decided to screen the optimal reaction conditions. The first investigation of copper salts showed that the copper sources significantly affected the reactivity and enantioselectivity (entries 3–9). Among the copper salts tested, Cu (OAc)₂·H₂O proved to be the best choice, which gave the addition product in 89% yield with 98% ee (entry 5). Subsequent solvents screen results



SCHEME 1 Asymmetric addition of diethylzinc to benzaldehyde with ligand (*R*)-**L1** and (*R*)-**L2**.



SCHEME 2 Synthesis of chiral P,N-ligand (R)-L3a and L3b.

TABLE 1 Reaction conditions optimization

Entry	Ligand	Copper salt	Solvent	Yield (%) ^b	Ee (%) ^c
1	(R)-L3a	CuBr	toluene	88	96
2	(R)-L3b	CuBr	toluene	70	42
3	(R)-L3a	CuCl	toluene	89	97
4	(R)-L3a	CuI	toluene	69	97
5	(R)-L3a	Cu (OAc) ₂ ·H ₂ O	toluene	89	98
6	(R)-L3a	Cu (OTf) ₂	toluene	45	36
7	(R)-L3a	Cu (CH ₃ CN)ClO ₄	toluene	78	94
8	(R)-L3a	Cu (CH ₃ CN) BF ₄	toluene	70	91
9	(R)-L3a	Cu (OTf)·(C ₆ H ₆) _{0.5}	toluene	61	85
10	(R)-L3a	Cu (OAc) ₂ ·H ₂ O	THF	<10	nd
11	(R)-L3a	Cu (OAc) ₂ ·H ₂ O	Et ₂ O	33	9
12	(R)-L3a	Cu (OAc) ₂ ·H ₂ O	CH ₂ Cl ₂	26	11
13 ^d	(R)-L3a	Cu (OAc) ₂ ·H ₂ O	toluene	86	98

^aReaction conditions: benzaldehyde **1a** (0.5 mmol), Et₂Zn (0.6 mmol, 1.2 eq), [Cu] (0.01 mmol, 2 mol%), (R)-L3 (0.0125 mmol, 2.5 mol%) in 3 ml solvent at 0 °C for 12 hr.

^bIsolated yields.

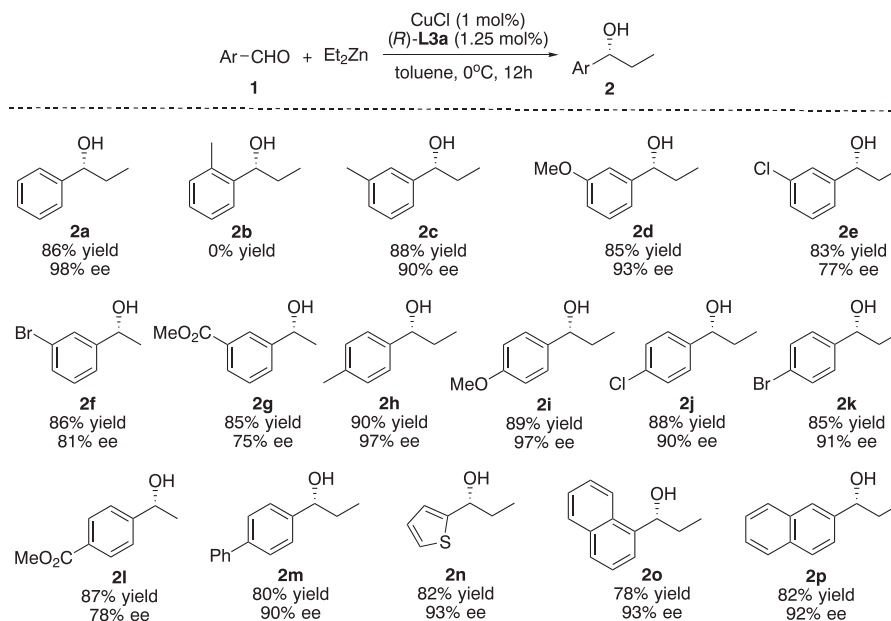
^cThe ee value was determined by HPLC.

^dS/C = 100.

displayed an obvious solvent dependency (entries 10–12). Only toluene was suitable solvent (entry 5), while other solvents such as THF, Et₂O and CH₂Cl₂ showed low reactivities and enantioselectivities (entries 10–12). Similar result was obtained when the catalyst loading was reduced to 1 mol% (S/C = 100, entry 13). Thus, the optimal reaction condition was identified as follow: Cu (OAc)₂·H₂O (1 mol%), ligand (R)-L3a (1.25 mol%) in toluene at 0 °C for 12 h.

Having established the optimal conditions, we explored various aromatic aldehydes to evaluate the scope and

limitation of the addition reaction. Scheme 3 summarized the results. A wide range of aromatic aldehydes reacted smoothly under the optimized conditions and generated the corresponding addition products in good to excellent enantioselectivities (75–98% ee). It appeared that the position of the substituent on the phenyl ring had significant effect on the reaction. The sterically hindered *ortho*-methyl benzaldehyde **1b** was not suitable substrate and no reaction occurred. In additional, the *para*-substituted substrates (**1h–1m**) tended to give higher enantioselectivities than the *meta*-substituted substrates (**1c–1g**). The



SCHEME 3 Substrate scope. Reaction conditions: aromatic aldehydes **1** (0.5 mmol), Et_2Zn (0.6 mmol, 1.2 eq), $\text{Cu}(\text{OAc})\cdot\text{H}_2\text{O}$ (0.005 mmol, 1 mol%), $(R)\text{-L3a}$ (0.00625 mmol, 1.25 mol%) in 3 ml toluene at 0°C for 12 hr. Yields of isolated product were given. The ee values were determined by HPLC and the absolute configuration was determined by comparison with the literatures.

electronic properties of the substituent group in the phenyl ring also had important effect on this transformation. Generally speaking, the substrates with electron-donating substituent (**1c-1d** and **1h-1i**) gave higher ee value than those with electron-withdraw substituent (**1e-1g** and **1j-1l**). A 4-phenylsubstituted benzaldehyde **1m** was also suitable for this reaction, giving the product **2m** with 90% ee. Meanwhile, the heteroaromatic aldehyde such as thiophene-2-carbaldehyde **1n** also worked well and generated the product **2n** with 93% ee. Furthermore, 1-naphthyl and 2-naphthyl substrates (**1o** and **1p**) also reacted smoothly, affording the corresponding products **2o** and **2p** with 93% and 92 ee, respectively.

3 | CONCLUSION

In conclusion, we have developed a new sterically hindered chiral P,N-ligand and successfully applied to the copper catalyzed asymmetric addition of diethylzinc to aromatic aldehydes. Under the optimized condition, a wide range of aromatic aldehydes can react smoothly, affording the corresponding addition products (chiral secondary alcohols) with good to excellent enantioselectivities. This reaction provides a readily accessible method for the synthesis of chiral secondary alcohols. Further applications of this new chiral P,N-ligand in asymmetric catalysis is still in progress.

4 | EXPERIMENTAL SECTION

4.1 | General methods

All reactions were carried out under N_2 atmosphere. Solvents were purified by standard procedure. Commercial reagents were used without further purification. ^1H NMR spectra were recorded on a Bruker 400 MHz spectrometer. Chemical shifts are expressed in δ value (ppm) using tetramethylsilane (TMS) as an internal standard. HPLC analysis was performed on an Agilent 1260 series instrument with a chiralpak AS-H, chiralcel OD-H, chiralcel AD-H or chiralcel OJ-H column. Copies of HPLC analysis and NMR spectra are given in the supporting information

4.2 | Procedure for synthesis of chiral P,N-ligand (R)-L3

To a stirred solution of amine **4** (453 mg, 1 mmol) in CH_2Cl_2 (10 ml) was added 6-phenyl-2-pyridinecarboxylic acid (797 mg, 4 mmol), 1,3-dicyclohexylcarbodiimide (DCC, 1.07 g, 5.2 mmol) and 4-(dimethylamino)pyridine (DMAP, 24 mg, 0.2 mmol). The result mixture was stirred overnight at room temperature. Water (10 ml) and AcOH (0.1 ml) was added and the mixture was stirred for 2 hrs. The solid residue was then filtered and washed with CH_2Cl_2 . The organic layer was separated, dried over

anhydrous Na_2SO_4 and concentrated. The residue was purified by flash column chromatography to afford 530 mg (84% yield) of (*R*)-**L3a** as a white solid. $[\alpha]_{\text{D}}^{20} = -32.1$ (*c* 1.10, CHCl_3). ^1H NMR (400 MHz, CDCl_3) δ 9.74 (s, 1H), 9.11 (d, *J* = 9.1 Hz, 1H), 8.15–8.01 (m, 4H), 7.93 (d, *J* = 8.1 Hz, 1H), 7.84 (t, *J* = 7.7 Hz, 1H), 7.77 (dd, *J* = 7.9, 1.0 Hz, 1H), 7.57–7.51 (m, 2H), 7.44–7.30 (m, 6H), 7.22–7.09 (m, 6H), 7.07–7.00 (m, 4H), 6.87–6.81 (m, 2H), 6.71–6.63 (m, 2H). ^{31}P NMR (162 MHz, CDCl_3) δ -14.06. ^{13}C NMR (100 MHz, CDCl_3) δ 161.9, 155.2, 149.6, 140.3, 140.0, 138.3, 138.2, 138.1, 137.5, 137.4, 137.3, 136.4 (d, *J* = 12.6 Hz), 135.0 (d, *J* = 2.7 Hz), 133.9, 133.8, 133.6, 133.5, 133.4, 133.3, 133.2, 132.9 (d, *J* = 6.9 Hz), 130.6 (d, *J* = 18.2 Hz), 129.4, 129.3, 129.1, 128.8, 128.3, 128.2, 128.1, 128.0, 127.9, 127.4 (d, *J* = 4.2 Hz), 126.6, 126.4, 126.3, 125.8, 124.7, 124.2 (d, *J* = 8.5 Hz), 122.8, 120.8, 119.5. HRMS calcd for $\text{C}_{44}\text{H}_{32}\text{N}_2\text{OP}$ [*M* + *H*] $^+$: 635.2252, found: 635.2259.

Ligand (*R*)-**L3b** was obtained as light yellow solid with 82% yield. $[\alpha]_{\text{D}}^{20} = -27.1$ (*c* 1.29, CHCl_3). ^1H NMR (400 MHz, CDCl_3) δ 10.16 (s, 1H), 9.05 (d, *J* = 9.0 Hz, 1H), 8.26–8.10 (m, 4H), 8.02 (d, *J* = 8.2 Hz, 1H), 7.94 (d, *J* = 8.2 Hz, 1H), 7.83 (d, *J* = 8.2 Hz, 1H), 7.72–7.65 (m, 2H), 7.58 (t, *J* = 7.4 Hz, 1H), 7.50 (t, *J* = 6.9 Hz, 1H), 7.37 (t, *J* = 7.5 Hz, 1H), 7.32–7.28 (m, 3H), 7.24–7.07 (m, 8H), 7.00 (d, *J* = 8.4 Hz, 1H), 6.89 (t, *J* = 7.5 Hz, 1H), 6.66 (t, *J* = 7.5 Hz, 2H). ^{31}P NMR (162 MHz, CDCl_3) δ -13.77. ^{13}C NMR (100 MHz, CDCl_3) 161.3, 149.4, 145.7, 140.4, 140.0, 138.2 (d, *J* = 13.4 Hz), 137.1, 137.0, 136.7, 136.6, 134.8 (d, *J* = 2.4 Hz), 134.2, 133.7, 133.5, 133.2 (d, *J* = 2.3 Hz), 133.0 (d, *J* = 6.9 Hz), 130.5, 129.9, 129.5, 129.1, 128.8, 128.3, 128.2, 128.1, 128.0, 127.9, 127.8, 127.5, 127.2 (d, *J* = 7.3 Hz), 126.3, 125.8, 124.5, 124.0 (d, *J* = 8.3 Hz), 118.8, 118.3. HRMS calcd for $\text{C}_{42}\text{H}_{30}\text{N}_2\text{OP}$ [*M* + *H*] $^+$: 609.2096, found: 609.2089.

4.3 | General procedure for asymmetric addition of Et_2Zn to aromatic aldehydes

The chiral P,N-ligand (*R*)-**L3a** (3.97 mg, 0.00625 mmol), $\text{Cu}(\text{OAc})_2 \cdot \text{H}_2\text{O}$ (1.00 mg, 0.005 mmol) and dry toluene (1 ml) was added to a Schlenk tube under nitrogen. After stirring for 1 hr at room temperature, a solution of the aryl aldehydes **1** (0.5 mmol) in 1 ml of toluene was added. The reaction mixture was stirred at room temperature for 10 min and then cooled to 0 °C. Et_2Zn (1.0 M in hexane, 0.6 ml, 0.6 mmol, 1.2 eq) was added slowly. The resulting mixture was further stirred for 12 hrs 0 °C. The solution was diluted with diethyl ether and washed successively with 1 N HCl and brine. The organic layer was dried over anhydrous Na_2SO_4 . The solvent was removed to give the crude product which was purified by flash column

chromatography to give the corresponding product. The enantiomeric excesses were determined by HPLC on a Chiralcel column.

4.3.1 | (*R*)-1-phenylpropan-1-ol (**2a**) [^{9b}, ^{10a}]

The title compound was prepared according to the General Procedure in 86% yield and 98% ee [Determined by HPLC with a chiralcel OJ-H column, hexane/isopropanol = 95/5, 0.8 ml/min, 210 nm, 40 °C; *t* (minor) = 9.4 min, *t* (major) = 9.9 min]. $[\alpha]_{\text{D}}^{20} = +10.9$ (*c* 0.8, CH_2Cl_2). ^1H NMR (400 MHz, CDCl_3) δ 7.43–7.26 (m, 5H), 4.62 (t, *J* = 6.6 Hz, 1H), 1.92–1.71 (m, 3H), 0.95 (t, *J* = 7.4 Hz, 3H).

4.3.2 | (*R*)-1-(*m*-tolyl)propan-1-ol (**2c**) [^{9b}]

The title compound was prepared according to the General Procedure in 88% yield and 90% ee [Determined by HPLC with a chiralcel OD-H column, hexane/isopropanol = 98/2, 0.8 ml/min, 210 nm, 40 °C; *t* (major) = 14.1 min, *t* (minor) = 16.1 min]. $[\alpha]_{\text{D}}^{20} = +20.9$ (*c* 0.97, CH_2Cl_2). ^1H NMR (400 MHz, CDCl_3) δ 7.28–7.24 (m, 1H), 7.22–7.08 (m, 3H), 4.58 (t, *J* = 6.6 Hz, 1H), 2.39 (s, 3H), 1.92 (s, 1H), 1.89–1.72 (m, 2H), 0.95 (t, *J* = 7.4 Hz, 3H).

4.3.3 | (*R*)-1-(3-methoxyphenyl)propan-1-ol (**2d**) [^{9b}, ^c]

The title compound was prepared according to the General Procedure in 85% yield and 93% ee [Determined by HPLC with a chiralpak AS-H column, hexane/isopropanol = 98/2, 0.8 ml/min, 210 nm, 40 °C; *t* (major) = 20.9 min, *t* (minor) = 23.1 min]. $[\alpha]_{\text{D}}^{20} = +18.1$ (*c* 1.14, CH_2Cl_2). ^1H NMR (400 MHz, CDCl_3) δ 7.31–7.26 (m, 2H), 6.98–6.91 (m, 2H), 6.88–6.80 (m, 1H), 4.60 (t, *J* = 6.6 Hz, 1H), 3.84 (s, 3H), 1.88–1.74 (m, 3H), 0.95 (t, *J* = 7.4 Hz, 3H).

4.3.4 | (*R*)-1-(3-chlorophenyl)propan-1-ol (**2e**) [^{9b}]

The title compound was prepared according to the General Procedure in 83% yield and 77% ee [Determined by HPLC with a chiralcel OJ-H column, hexane/isopropanol = 98/2, 0.8 ml/min, 210 nm, 40 °C; *t* (minor) = 16.0 min, *t* (major) = 17.9 min]. $[\alpha]_{\text{D}}^{20} = +11.8$ (*c* 1.09, CH_2Cl_2). ^1H NMR (400 MHz, CDCl_3) δ 7.39–7.35 (m, 1H), 7.33–7.28 (m, 1H), 7.28–7.21 (m, 2H), 4.61 (t, *J* = 6.5 Hz, 1H), 1.88–1.72 (m, 3H), 0.94 (t, *J* = 7.4 Hz, 3H).

4.3.5 | (R)-1-(3-bromophenyl)propan-1-ol (2f) ^[9c]

The title compound was prepared according to the General Procedure in 86% yield and 81% ee [Determined by HPLC with a chiralcel OJ-H column, hexane/isopropanol = 90/10, 0.8 ml/min, 210 nm, 40 °C; t (minor) = 18.1 min, t (major) = 21.4 min]. $[\alpha]_{\text{D}}^{20} = +11.4$ (c 0.95, CH₂Cl₂). ¹H NMR (400 MHz, CDCl₃) δ 7.55–7.50 (m, 1H), 7.45–7.39 (m, 1H), 7.29–7.20 (m, 2H), 4.59 (t, *J* = 6.5 Hz, 1H), 1.96 (s, 1H), 1.86–1.68 (m, 2H), 0.94 (t, *J* = 7.4 Hz, 3H).

4.3.6 | Methyl (R)-3-(1-hydroxypropyl)benzoate (2 g) ^[15]

The title compound was prepared according to the General Procedure in 85% yield and 75% ee [Determined by HPLC with a chiralcel AD-H column, hexane/isopropanol = 90/10, 0.8 ml/min, 210 nm, 40 °C; t (major) = 10.8 min, t (minor) = 11.9 min]. $[\alpha]_{\text{D}}^{20} = +27.8$ (c 1.05, CH₂Cl₂). ¹H NMR (400 MHz, CDCl₃) δ 8.01 (s, 1H), 7.94 (d, *J* = 7.7 Hz, 1H), 7.55 (d, *J* = 7.6 Hz, 1H), 7.42 (t, *J* = 7.7 Hz, 1H), 4.67 (t, *J* = 6.5 Hz, 1H), 3.91 (s, 3H), 1.92 (s, 1H), 1.87–1.73 (m, 2H), 0.92 (t, *J* = 7.4 Hz, 3H).

4.3.7 | (R)-1-(*p*-tolyl)propan-1-ol (2 h) ^[9c]

The title compound was prepared according to the General Procedure in 90% yield and 97% ee [Determined by HPLC with a chiralcel AD-H column, hexane/isopropanol = 90/10, 0.8 ml/min, 210 nm, 40 °C; t (major) = 6.5 min, t (minor) = 7.1 min]. $[\alpha]_{\text{D}}^{20} = +9.9$ (c 1.13, CH₂Cl₂). ¹H NMR (400 MHz, CDCl₃) δ 7.26 (d, *J* = 8.0 Hz, 2H), 7.19 (d, *J* = 8.0 Hz, 2H), 4.58 (t, *J* = 6.6 Hz, 1H), 2.38 (s, 3H), 1.93 (br, 1H), 1.90–1.72 (m, 2H), 0.94 (t, *J* = 7.4 Hz, 3H).

4.3.8 | (R)-1-(4-methoxyphenyl)propan-1-ol (2i) ^[9c, 10a]

The title compound was prepared according to the General Procedure in 89% yield and 97% ee [Determined by HPLC with a chiralpak AD-H column, hexane/isopropanol = 90/10, 0.8 ml/min, 210 nm, 40 °C; t (major) = 8.8 min, t (minor) = 9.6 min]. $[\alpha]_{\text{D}}^{20} = +26.4$ (c 1.09, CH₂Cl₂). ¹H NMR (400 MHz, CDCl₃) δ 7.29 (d, *J* = 8.6 Hz, 2H), 6.91 (d, *J* = 8.6 Hz, 2H), 4.56 (t, *J* = 6.7 Hz, 1H), 3.83 (s, 3H), 1.90–1.68 (m, 3H), 0.92 (t, *J* = 7.4 Hz, 3H).

4.3.9 | (R)-1-(4-chlorophenyl)propan-1-ol (2j) ^[9b, 10a]

The title compound was prepared according to the General Procedure in 88% yield and 90% ee [Determined by HPLC with a chiralcel OJ-H column, hexane/isopropanol = 97/3, 0.8 ml/min, 210 nm, 40 °C; t (major) = 12.6 min, t (minor) = 13.1 min]. $[\alpha]_{\text{D}}^{20} = +24.5$ (c 1.10, CH₂Cl₂). ¹H NMR (400 MHz, CDCl₃) δ 7.38–7.25 (m, 4H), 4.60 (t, *J* = 6.6 Hz, 1H), 1.92 (br, 1H), 1.86–1.68 (m, 2H), 0.92 (t, *J* = 7.4 Hz, 3H).

4.3.10 | (R)-1-(4-bromophenyl)propan-1-ol (2 k) ^[9c]

The title compound was prepared according to the General Procedure in 85% yield and 91% ee [Determined by HPLC with a chiralpak AS-H column, hexane/isopropanol = 98/2, 0.8 ml/min, 210 nm, 40 °C; t (major) = 17.3 min, t (minor) = 18.8 min]. $[\alpha]_{\text{D}}^{20} = +16.0$ (c 1.10, CH₂Cl₂). ¹H NMR (400 MHz, CDCl₃) δ 7.49 (d, *J* = 8.4 Hz, 2H), 7.23 (d, *J* = 8.4 Hz, 2H), 4.59 (t, *J* = 6.5 Hz, 1H), 1.93 (s, 1H), 1.86–1.67 (m, 2H), 0.92 (t, *J* = 7.4 Hz, 3H).

4.3.11 | Methyl (R)-4-(1-hydroxypropyl)benzoate (2 l) ^[16]

The title compound was prepared according to the General Procedure in 87% yield and 78% ee [Determined by HPLC with a chiralcel AD-H column, hexane/isopropanol = 90/10, 0.8 ml/min, 210 nm, 40 °C; t (major) = 9.3 min, t (minor) = 10.1 min]. $[\alpha]_{\text{D}}^{20} = +24.4$ (c 1.09, CH₂Cl₂). ¹H NMR (400 MHz, CDCl₃) δ 8.02 (d, *J* = 8.0 Hz, 2H), 7.42 (d, *J* = 8.1 Hz, 2H), 4.69 (t, *J* = 6.5 Hz, 1H), 3.92 (s, 3H), 2.11 (br, 1H), 1.88–1.71 (m, 2H), 0.94 (t, *J* = 7.4 Hz, 3H).

4.3.12 | (R)-1-([1,1'-biphenyl]-4-yl)propan-1-ol (2 m) ^[17]

The title compound was prepared according to the General Procedure in 80% yield and 90% ee [Determined by HPLC with a chiralcel AD-H column, hexane/isopropanol = 95/5, 0.8 ml/min, 210 nm, 40 °C; t (minor) = 13.2 min, t (major) = 14.7 min]. $[\alpha]_{\text{D}}^{20} = +24.5$ (c 1.06, CH₂Cl₂). ¹H NMR (400 MHz, CDCl₃) δ 7.70–7.60 (m, 4H), 7.55–7.36 (m, 5H), 4.67 (t, *J* = 6.6 Hz, 1H), 2.34 (br, 1H), 1.97–1.76 (m, 2H), 1.01 (t, *J* = 7.4 Hz, 3H).

4.3.13 | (R)-1-(thiophen-2-yl)propan-1-ol (2n) [9b, 10a]

The title compound was prepared according to the General Procedure in 82% yield and 93% ee [Determined by HPLC with a chiralcel OJ-H column, hexane/isopropanol = 90/10, 0.8 ml/min, 210 nm, 40 °C; t (minor) = 7.6 min, t (major) = 8.6 min]. $[\alpha]_D^{20} = +13.5$ (c 0.82, CH₂Cl₂). ¹H NMR (400 MHz, CDCl₃) δ 7.28–7.25 (m, 1H), 7.03–6.96 (m, 2H), 4.87 (t, *J* = 6.6 Hz, 1H), 1.96–1.86 (m, 3H), 0.99 (t, *J* = 7.4 Hz, 3H).

4.3.14 | (R)-1-(naphthalen-1-yl)propan-1-ol (2o) [9b, 10a]

The title compound was prepared according to the General Procedure in 78% yield and 93% ee [Determined by HPLC with a chiralcel AD-H column, hexane/isopropanol = 95/5, 0.8 ml/min, 210 nm, 40 °C; t (minor) = 12.8 min, t (major) = 14.1 min]. $[\alpha]_D^{20} = +75.2$ (c 0.79, CH₂Cl₂). ¹H NMR (400 MHz, CDCl₃) δ 8.18–8.10 (m, 1H), 7.94–7.88 (m, 1H), 7.81 (d, *J* = 8.2 Hz, 1H), 7.66 (d, *J* = 7.1 Hz, 1H), 7.59–7.47 (m, 3H), 5.41 (dd, *J* = 7.4, 5.1 Hz, 1H), 2.15 (s, 1H), 2.09–1.90 (m, 2H), 1.06 (t, *J* = 7.4 Hz, 3H).

4.3.15 | (R)-1-(naphthalen-2-yl)propan-1-ol (2p) [9b, 10a]

The title compound was prepared according to the General Procedure in 82% yield and 92% ee [Determined by HPLC with a chiralpak AS-H column, hexane/isopropanol = 98/2, 0.8 ml/min, 210 nm, 40 °C; t (major) = 15.0 min, t (minor) = 18.4 min]. $[\alpha]_D^{20} = +27.3$ (c 1.07, CH₂Cl₂). ¹H NMR (400 MHz, CDCl₃) δ 7.90–7.82 (m, 3H), 7.79 (s, 1H), 7.56–7.46 (m, 3H), 4.76 (t, *J* = 6.6 Hz, 1H), 2.30 (s, 1H), 1.99–1.81 (m, 2H), 0.97 (t, *J* = 7.4 Hz, 3H).

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